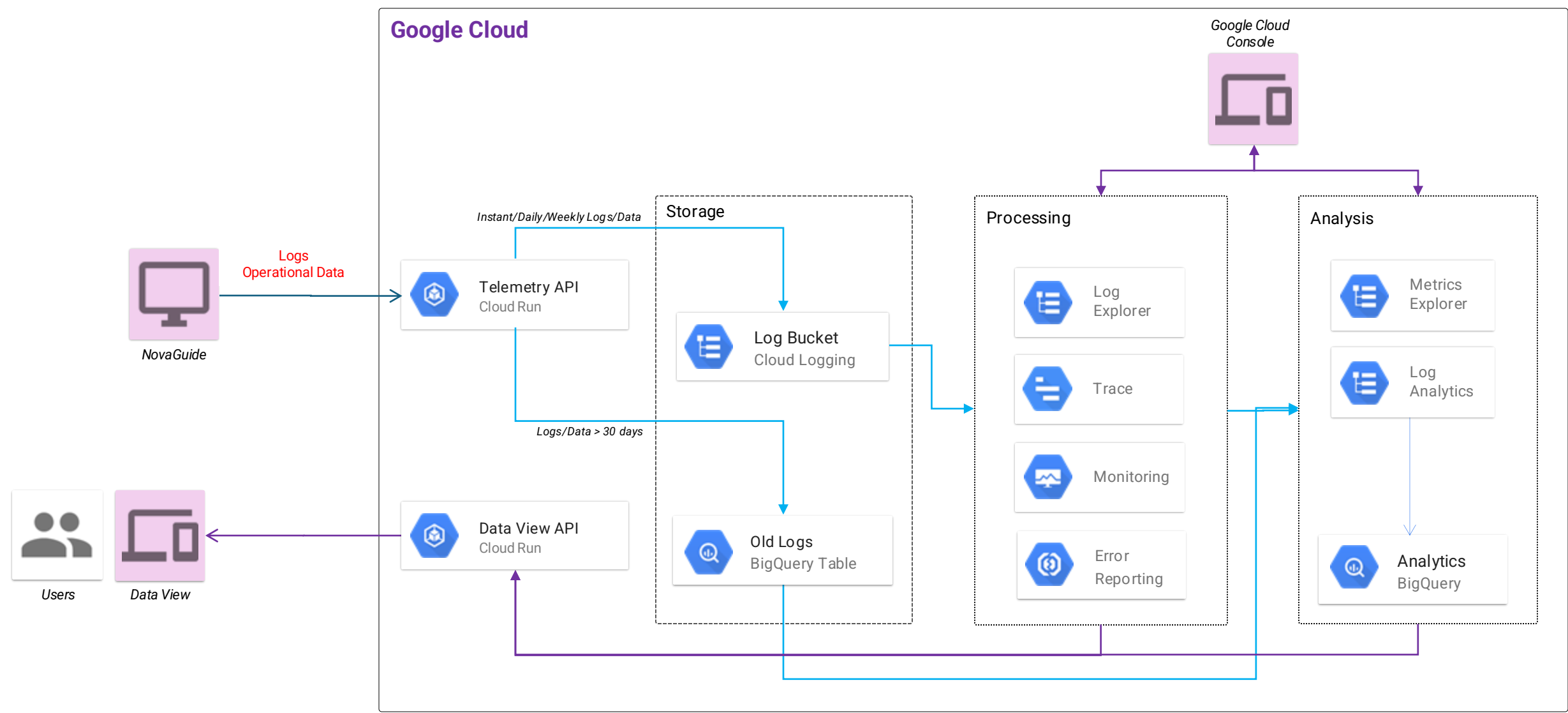


# Centralized Telemetry

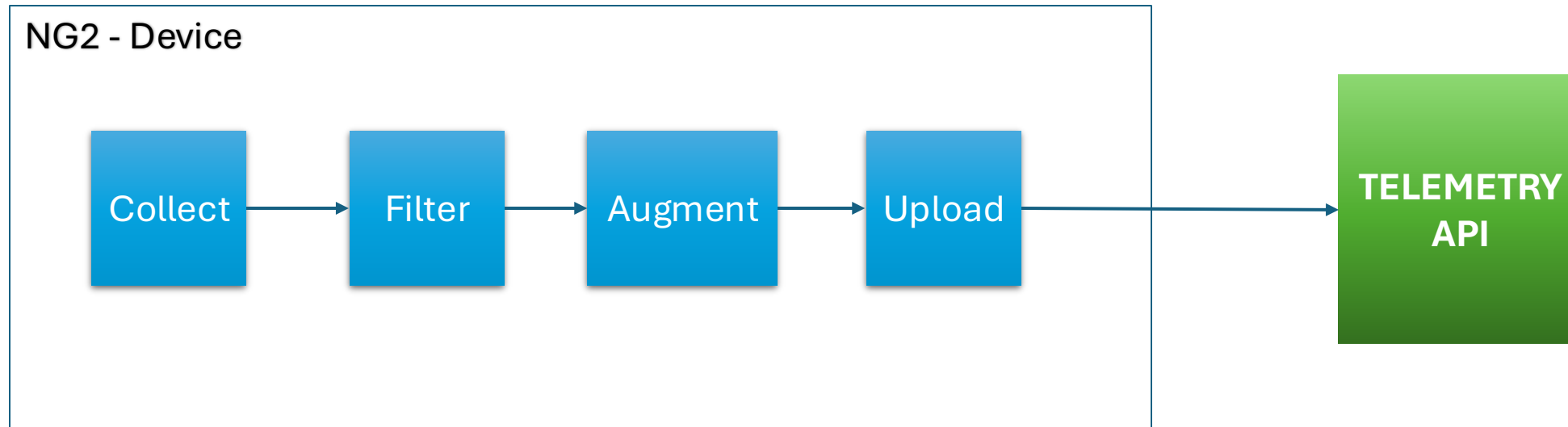
# Introduction

- Objective – Create an application to investigate the logs and figure out how to minimize the maintenance after it happened and preferably do preventative maintenance
- Status – PoC
  - NG2 has a tool to take all the logs created in the system and analyze it using wild cards.
  - Planning to do a PoC to move the logs to the cloud and figure out useful insights
- Issues –
  - Existing logs uploaded to the cloud concern on PHI
  - Draw insights from the logs that are available
- Recommendations
  - Remove PHI at the source and send the operational data alone
  - Leverage AI in the cloud

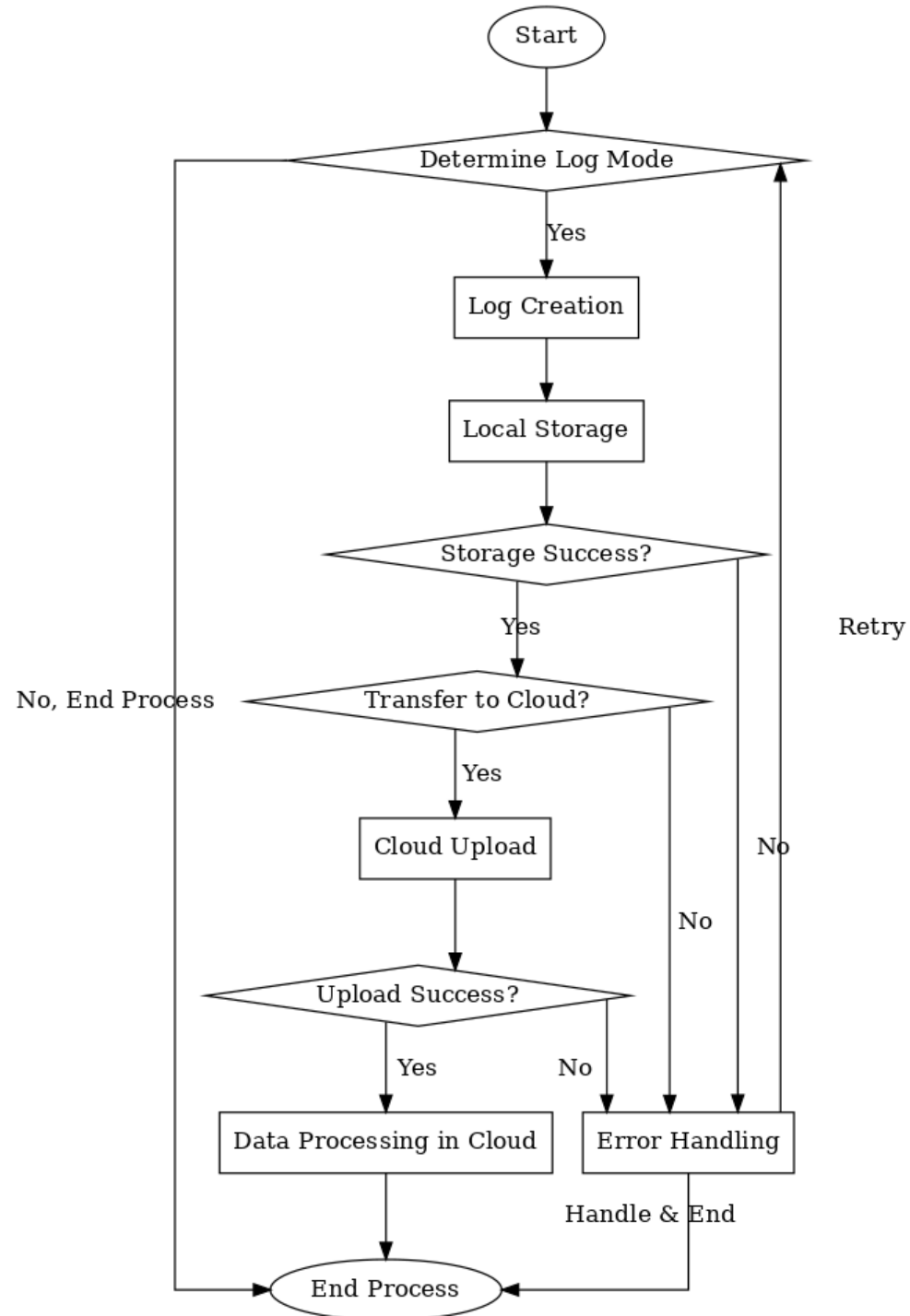
# NeuraSignal Telemetry Architecture



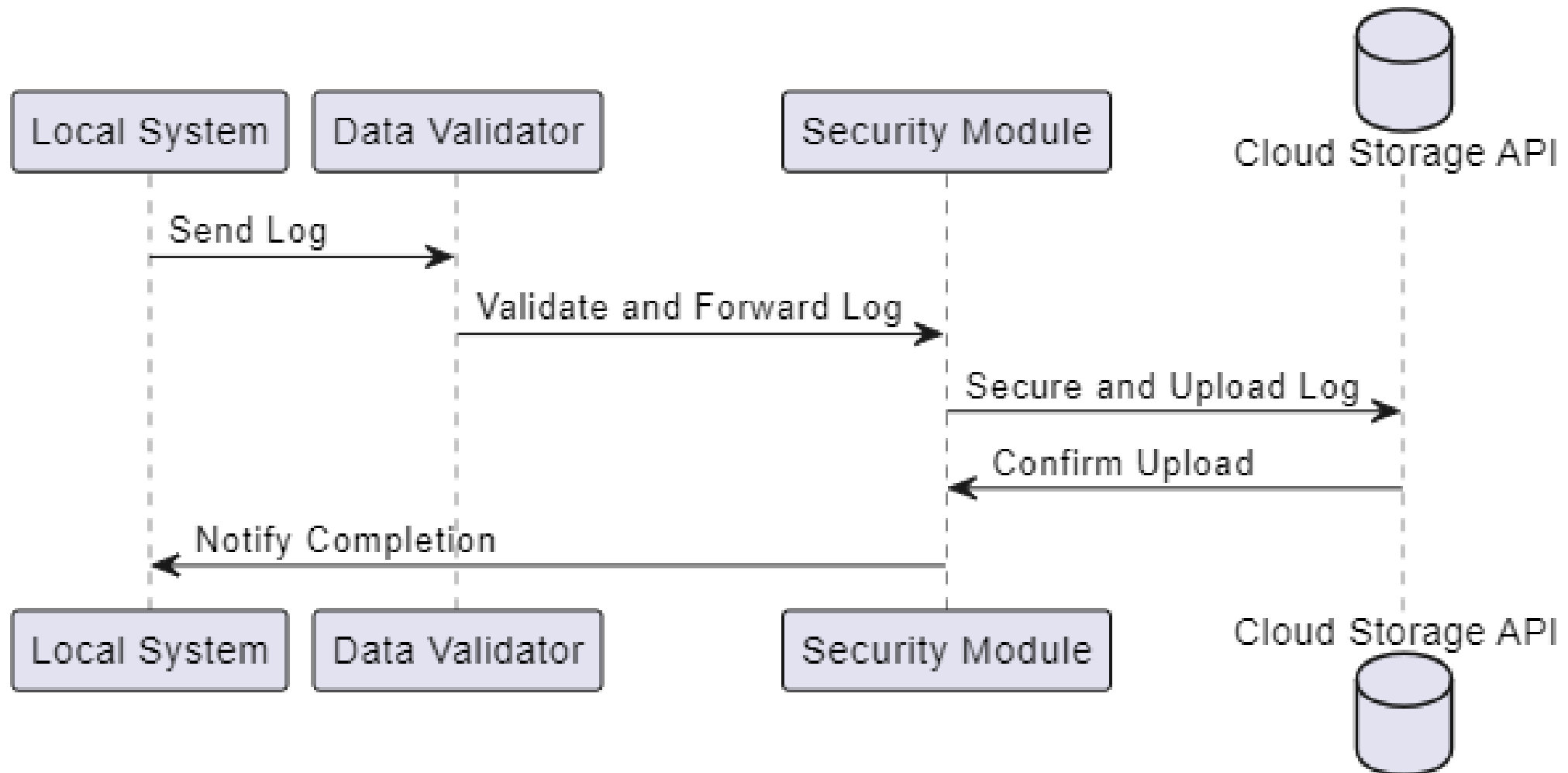
# NG2



# Workflow



# Message Sequence Chart (MSC)



Ingest

# Options

- **SDKs or APIs**
- Log Forwarding Agents – Fluentd
- **Cloud-Native Log Management Services** – Google Cloud Logging
- ELK
- Serverless – Google Cloud Function



# Processing

# Options

- BigQuery
- **Cloud Logging**
- Dataflow – logs need to be processed in real-time or batch
- Pub/Sub – real-time log data processing
- Looker – BI and operational dashboard

# Analysis & Monitoring

# Options

- **Google Cloud Logging**
- Looker
- VueJS Frontend
- PowerBI
- Third Party – Fluentd